

System and Network Security

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Elliptic Curve Cryptography (ECC)

Elliptic Curve Cryptography (ECC)

- ECC makes use of the elliptic curves (not ellipses) in which the variables and coefficients are all restricted to elements of a finite field.
- Two family of elliptic curves are used in ECC:
 - prime curves defined over Z_p , that is, $GF(p)$, p being a prime.
 - binary curves constructed over $GF(2^n)$.

Elliptic curves over the reals

Definition

Let $a, b \in R$ be constants such that $4a^3 + 27b^2 \neq 0$. A non-singular elliptic curve is the set E of solutions $(x, y) \in R \times R$ to the equation

$$y^2 = x^3 + ax + b,$$

together with a special point $\mathcal{O}$ called the point at infinity (or zero point).

Elliptic curves over the reals

- It can be shown that the condition $4a^3 + 27b^2 \neq 0$ is the necessary and sufficient to ensure that the equation $x^3 + ax + b = 0$ has three distinct roots (may be real or complex numbers) (by Cardan Method).
- If $4a^3 + 27b^2 = 0$, the corresponding elliptic curve is called singular.
- If $P = (x_P, y_P)$ and $Q = (x_Q, y_Q)$, then $P + Q = \mathcal{O}$ implies that $x_Q = x_P$ and $y_Q = -y_P$.
- Also, $P + \mathcal{O} = \mathcal{O} + P = P$ for all $P \in E$.

Elliptic Curve Cryptography (ECC)

Elliptic curves over modulo a prime $GF(p)$

Definition

Let $p > 3$ be a prime. The elliptic curve $y^2 = x^3 + ax + b$ over Z_p is the set $E_p(a, b)$ of solutions $(x, y) \in E_p(a, b)$ to the congruence

$$y^2 = x^3 + ax + b \pmod{p},$$

where $a, b \in Z_p$ are constants such that $4a^3 + 27b^2 \not\equiv 0 \pmod{p}$, together with a special point $\mathcal{O}$ called the point at infinity (or zero point).

Elliptic curves over modulo a prime $GF(p)$

Properties of Elliptic Curves

- An elliptic curve $E_p(a, b)$ over Z_p (p prime, $p > 3$) will have roughly p points on it.
- More precisely, a well-known theorem due to Hasse asserts that the number of points on $E_p(a, b)$, which is denoted by $\#E$, satisfies the following inequality:

$$p + 1 - 2\sqrt{p} \leq \#E \leq p + 1 + 2\sqrt{p}.$$

- In addition, $E_p(a, b)$ forms an abelian or commutative group under addition modulo p operation.

References

- N. Koblitz. Elliptic Curve Cryptosystems. Mathematics of Computation, Vol. 48, pp. 203-209, 1987.
- V. Miller. Uses of elliptic curves in cryptography. Advances in Cryptology - CRYPTO'85, Lecture Notes in Computer Science (LNCS), Springer, Vol. 218, pp. 417-426, 1986.
- Douglas R. Stinson. Cryptography: Theory and Practice, Chapman & Hall/CRC, 2nd Edition, 2005.

Elliptic curves over modulo a prime $GF(p)$

Finding an inverse

- The inverse of a point $P = (x_P, y_P) \in E_p(a, b)$ is $-P = (x_P, -y_P)$, where $-y$ is the additive inverse of y .
- For example, if $p = 13$, the inverse of $(4, 2)$ is $(4, 11)$.

Elliptic Curve Cryptography (ECC)

Elliptic curves over modulo a prime $GF(p)$

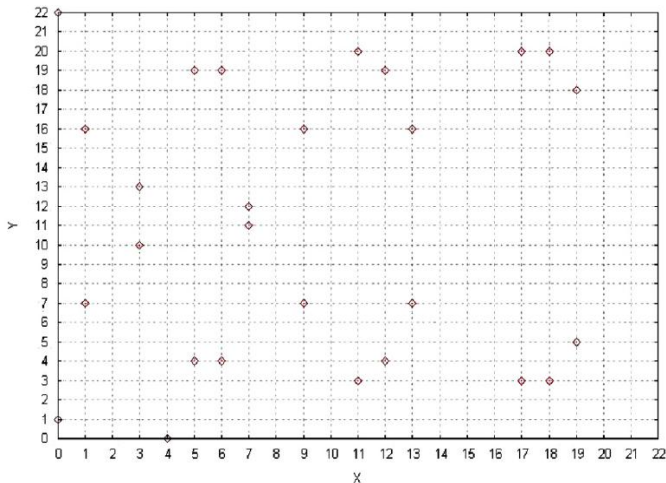
Finding all points on an elliptic curve

Algorithm: EllipticCurvePoints (p, a, b)

```
1:  $x \leftarrow 0$ 
2: while  $x < p$  do
3:    $w \leftarrow (x^3 + ax + b) \pmod{p}$  /*  $w = y^2$  */
4:   if ( $w$  is a perfect square in  $Z_p$ ) then
5:     Output  $(x, \sqrt{w}), (x, -\sqrt{w})$ 
6:   end if
7:    $x \leftarrow x + 1$ 
8: end while
```

Elliptic Curve Cryptography (ECC)

Example of elliptic curve in case of $y^2 = x^3 + x + 1 \pmod{23}$.



Elliptic Curve Cryptography (ECC)

Point addition on elliptic curve over finite field $GF(p)$

Let G be the base point on $E_p(a, b)$ whose order be n , that is, $nG = G + G + \dots + G$ (n times) $= \mathcal{O}$.

If $P = (x_P, y_P)$ and $Q = (x_Q, y_Q)$ be two points on elliptic curve $y^2 = x^3 + ax + b \pmod{p}$, $R = (x_R, y_R) = P + Q$ is computed as follows:

$$\begin{aligned}x_R &= (\lambda^2 - x_P - x_Q) \pmod{p}, \\y_R &= (\lambda(x_P - x_R) - y_P) \pmod{p}, \\ \text{where } \lambda &= \begin{cases} \frac{y_Q - y_P}{x_Q - x_P} \pmod{p}, & \text{if } P \neq -Q \\ \frac{3x_P^2 + a}{2y_P} \pmod{p}, & \text{if } P = Q. \end{cases}\end{aligned}$$

Elliptic Curve Cryptography (ECC)

Scalar multiplication on elliptic curve over finite field $GF(p)$

If $P = (x_P, y_P)$ be a point on elliptic curve $y^2 = x^3 + ax + b \pmod{p}$, then then $5P$ is computed as $5P = P + P + P + P + P$.

Think about optimization method?

Reference: N Tiwari, S Padhye. Provable Secure Multi-Proxy Signature Scheme without Bilinear Maps. International Journal of Network Security, Vol. 17, No. 1, pp. 288-293, 2015.

Elliptic Curve Cryptography (ECC)

Problem: Consider two points $P = (11, 3)$ and $Q = (9, 7)$ in the elliptic curve $E_{23}(1, 1)$. Compute $P + Q$ and $2P$.

In order to compute $R = P + Q = (x_R, y_R)$, we first compute λ as

$$\begin{aligned}\lambda &= \frac{7 - 3}{9 - 11} \pmod{23} \\ &= -2 \pmod{23} \\ &= 21.\end{aligned}\tag{1}$$

Thus, x_R and y_R are derived as

$$\begin{aligned}x_R &= (21^2 - 11 - 9) \pmod{23} = 7, \\ y_R &= (21(11 - 7) - 3) \pmod{23} = 12.\end{aligned}$$

As a result, $P + Q = (7, 12)$.

Elliptic Curve Cryptography (ECC)

Problem: Consider two points $P = (11, 3)$ and $Q = (9, 7)$ in the elliptic curve $E_{23}(1, 1)$. Compute $P + Q$ and $2P$.

In order to compute $R = 2P = (x_R, y_R)$, we must first derive λ as follows:

$$\lambda = \frac{3(11^2) + 1}{2 \times 3} \pmod{23} = 7.$$

Hence, $R = P + P = (x_R, y_R)$ is computed as

$$x_R = (7^2 - 11 - 11) \pmod{23} = 4,$$

$$y_R = (7(11 - 4) - 3) \pmod{23} = 0,$$

and, thus $2P = (4, 0)$.

Elliptic Curve Discrete Logarithm Problem (ECDLP)

- Let $E_p(a, b)$ be an elliptic curve modulo a prime p .
- Given two points $P \in E_p(a, b)$ and $Q = kP \in E_p(a, b)$, for some positive integer k , where $Q = kP$ represent the point P on elliptic curve $E_p(a, b)$ be added to itself k times.
- Then the elliptic curve discrete logarithm problem (ECDLP) is to determine k given P and Q .
- It is computationally easy to calculate Q given k and P , but it is computationally infeasible to determine k given Q and P , when the prime p is large.

Elliptic Curve Cryptography (ECC)

Elliptic Curve Discrete Logarithm Problem (ECDLP)

Definition

Let $E_p(a, b)$ be an elliptic curve modulo a prime p , and $P \in E_p(a, b)$ and $Q = kP \in E_p(a, b)$ be two points, where $k \in_R Z_p$ (We use the notation $a \in_R B$ to denote that a is randomly chosen from the set B).

Instance: (P, Q, m) for some $k, m \in_R Z_p$.

Output: **Yes**, if $Q = mP$, i.e., $k = m$, and **No**, otherwise.

Consider the following two probability distributions:

$$D_{real} = \{k \in_R Z_p, U = P, V = Q(= kP), W = k : (U, V, W)\}, \text{ and}$$

$$D_{rand} = \{k, m \in_R Z_p, U = P, V = Q(= kP), W = m : (U, V, W)\}.$$

Elliptic Curve Cryptography (ECC)

Elliptic Curve Discrete Logarithm Problem (ECDLP)

Definition

The advantage of any probabilistic polynomial-time (PPT), 0/1-valued distinguisher $\mathcal{D}$ in solving *ECDLP* on $E_p(a, b)$ is defined as

$$\text{Adv}_{\mathcal{D}, E_p(a, b)}^{\text{ECDLP}} = |Pr[(U, V, W) \leftarrow D_{\text{real}} : \mathcal{D}(U, V, W) = 1] - Pr[(U, V, W) \leftarrow D_{\text{rand}} : \mathcal{D}(U, V, W) = 1]|,$$

where the probability $Pr[\cdot]$ is taken over the random choices of k and m . $\mathcal{D}$ is called an (t, ϵ) -ECDLP distinguisher for $E_p(a, b)$ if $\mathcal{D}$ runs at most in time t with $\text{Adv}_{\mathcal{D}, E_p(a, b)}^{\text{ECDLP}}(t) \geq \epsilon$.

ECDLP assumption: There exists no (t, ϵ) -ECDLP distinguisher for $E_p(a, b)$. Thus, for every $\mathcal{D}$, $\text{Adv}_{\mathcal{D}, E_p(a, b)}^{\text{ECDLP}}(t) \leq \epsilon$, for any sufficiently small $\epsilon > 0$.

Elliptic Curve Cryptography (ECC)

Elliptic Curve Discrete Logarithm Problem (ECDLP)

In other words, ECDLP can be also formally defined as follows. For any PPT algorithm, say A (in the security parameter l), $\Pr[A(P, Q) = k] < \epsilon(l)$, where $\epsilon(l)$ is a negligible function depending on l .

References:

- Vanga Odelu, **Ashok Kumar Das**, and Adrijit Goswami. “A secure effective key management scheme for dynamic access control in a large leaf class hierarchy,” in **Information Sciences (Elsevier)**, Vol. 269, No. C, pp. 270-285, 2014. (2013 SCI Impact Factor: 3.893) [This article has been downloaded or viewed 484 times since publication during the period October 2013 to September 2014]
- **Ashok Kumar Das**, Nayan Ranjan Paul, and Laxminath Tripathy. “Cryptanalysis and improvement of an access control in user hierarchy based on elliptic curve cryptosystem,” in **Information Sciences (Elsevier)**, Vol. 209, No. C, pp. 80 - 92, 2012. (2013 SCI Impact Factor: 3.893)

Elliptic Curves over $GF(2^m)$

- In this the elliptic curve is of the form:

$$y^2 + xy = x^3 + ax^2 + b,$$

whose coefficients are in $GF(2^m)$ and the addition is modulo 2 ($\oplus$) and multiplication is AND operation.

- The rules for addition can be stated as follows.

For all points $P, Q \in E_{2^m}(a, b)$:

- $P + \mathcal{O} = \mathcal{O} + P = P$, where $\mathcal{O}$ is the point at infinity (or zero point).
- If $P = (x_P, y_P)$, then $P + (x_P, x_P + y_P) = \mathcal{O}$. Then $-P = (x_P, x_P + y_P)$.

Elliptic Curve Cryptography (ECC)

Elliptic Curves over $GF(2^m)$

- If $P = (x_P, y_P)$ and $Q = (x_Q, y_Q)$ with $P \neq -Q$ and $P \neq Q$, then $R = P + Q = (x_R, y_R)$ is defined by the following rules:

$$x_R = \lambda^2 + \lambda + x_P + x_Q + a,$$

$$y_R = \lambda(x_P + x_R) + x_R + y_P,$$

$$\lambda = \frac{y_Q + y_P}{x_Q + x_P} \pmod{2}.$$

- If $P = (x_P, y_P)$, then $R = P + P = (x_R, y_R)$ is defined by the following rules:

$$x_R = \lambda^2 + \lambda + a,$$

$$y_R = x_P^2 + (\lambda + 1)x_R,$$

$$\lambda = x_P + \frac{y_P}{x_P} \pmod{2}.$$

Elliptic Curve Cryptography (ECC)

Elliptic Curve Cryptography Key Exchange Protocol (ECC Key Exchange)

- Pick a large integer q , where $q = p$; p being a prime, or $q = 2^m$, for some positive integer m , and the elliptic curve parameters a and b for the elliptic curves:

$$\begin{aligned}y^2 &= x^3 + ax + b \pmod{p} \text{ in } GF(p); \\y^2 + xy &= x^3 + ax^2 + b \pmod{p} \text{ in } GF(2^m).\end{aligned}$$

- Pick a base point $G = (x, y)$ in $E_q(a, b)$ whose order is a very large value n , that is, $nG = \mathcal{O}$.
- $E_q(a, b)$ and G are parameters of the cryptosystem known to all participants.

Elliptic Curve Cryptography (ECC)

Elliptic Curve Cryptography Key Exchange Protocol (ECC Key Exchange)

A key exchange between two users A and B can be accomplished as follows:

- A selects an integer n_A , where $n_A < n$. A 's private key is n_A .
 A generates a public key $P_A = n_A G$; the public key is a point in $E_q(a, b)$.
- B similarly selects a private key n_B , where $n_B < n$. B 's private key is n_B .
 B generates a public key $P_B = n_B G$.
- A generates the secret key $K_{A,B} = n_A P_B$.
- B generates the secret key $K_{B,A} = n_B P_A$.

ECC Key Exchange Protocol (continued...)

Summary

User A	User B
1. Select private n_A 2. Calculate public P_A 3. $\underline{P_A = n_A G}$ →	1. Select private n_B 2. Calculate public P_B 3. $\underline{P_B = n_B G}$ ←
4. $K_{A,B} = n_A P_B$	4. $K_{B,A} = n_B P_A$

ECC Key Exchange Protocol (Continued...)

Correctness Proof

$$\begin{aligned}K_{A,B} &= n_A \times P_B [\text{User A}] \\&= n_A \times (n_B \times G) \\&= (n_A \times n_B) \times G \\&= (n_B \times n_A) \times G \\&= n_B \times (n_A \times G) \\&= n_B \times P_A \\&= K_{B,A} [\text{User B}]\end{aligned}$$

ECC Key Exchange Protocol (Continued...)

Problem [ECC Key Exchange]

Users A and B use the ECC key exchange technique with a common prime $q = 211$ and an elliptic curve $E_q(a, b)$, where $a = 0$ and $b = -4$. Let $G = (2, 2)$ a base point on $E_q(a, b)$.

(a) If user A has private key $n_A = 121$, what is the A 's public key P_A ?

Answer: $P_A = n_A \cdot G = 121 \cdot (2, 2) = (115, 48)$.

(b) If user B has private key $n_B = 203$, what is the B 's public key P_B ?

Answer: $P_B = n_B \cdot G = 203 \cdot (2, 2) = (130, 203)$.

(c) What is the secret shared key?

Answer: $K_{A,B} = n_A \cdot P_B = 121 \cdot (185, 178) = (161, 69)$, by user A .

$K_{B,A} = n_B \cdot P_A = 203 \cdot (67, 106) = (161, 69)$, by user B .

ECC Encryption

- The first task in this system is to encode the plaintext message m to be sent as an x-y point P_m in $E_q(a, b)$. (For example, Koblitz method (Available at <http://zoo.cs.yale.edu/classes/cs467/2012s/lectures/ln13.pdf>)).
- It is the point P_m that will be encrypted as a ciphertext and subsequently decrypted.
- As with the ECC key exchange system, an encryption/decryption system requires a base point G and an elliptic curve $E_q(a, b)$.
- Let user A 's private-public key pair $(KR_a, KU_a) = (n_A, P_A)$ and user B 's private-public key pair $(KR_b, KU_b) = (n_B, P_B)$

ECC Encryption

- To encrypt and send a plaintext message (encoded) P_m to user B , user A proceeds as follows:
 - A chooses a random positive integer k .
 - A produces the ciphertext C_m consisting of the pair of points

$$\begin{aligned}C_m &= E_{PU_b}(P_m) \\&= E_{P_B}(P_m) \\&= \{C_1, C_2\} \\&= \{kG, P_m + kP_B\},\end{aligned}$$

where $P_B = n_B \times G$.

ECC Decryption

- To decrypt the ciphertext C_m , user B proceeds as follows:
 - B uses its own private key $KR_b = n_B$.
 - The plaintext P_m is recovered as

$$\begin{aligned}P_m &= D_{PR_b}(C_m) \\&= D_{n_B}(C_m) \\&= C_2 - n_B \times C_1 \\&= P_m + kP_B - n_B \times (kG) \\&= P_m + k \times (n_B G) - n_B \times (kG) \\&= P_m.\end{aligned}$$

ECC Encryption/Decryption (Continued...)

Problem [ECC Encryption/Decryption]

Suppose two users A and B rely on the ECC key cryptosystem. An elliptic curve cryptosystem operates on the curve $y^2 = x^3 + ax + b \pmod{q}$ with the parameters $E_{11}(1, 6)$, and the base point $G = (2, 7)$. Assume that B 's private key is $n_B = 7$.

- (a) Find B 's public key P_B .
- (b) Determine the ciphertext C_m , if the user A sends the message $P_m = (10, 9)$ with the random value $k = 3$.

ECC Encryption/Decryption (Continued...)

Solution:

- (a) $P_B = n_B \times G = 7 \times (2, 7) = (7, 2)$.
- (b) $C_m = (C_1, C_2)$, where

$$\begin{aligned}C_1 &= k \times G \\&= 3 \times (2, 7) \\&= (8, 3),\end{aligned}$$

and

$$\begin{aligned}C_2 &= P_m + kP_B \\&= (10, 9) + 3 \times (7, 2) \\&= (10, 9) + (3, 5) \\&= (10, 2).\end{aligned}$$

Security of ECC

- The security of ECC depends on how difficult it is to determine the discrete logarithm k given kP and P . This is referred to as the “elliptic curve discrete logarithm problem (ECDLP)”. The fastest known technique for taking the elliptic curve logarithm is known as the Pollard rho method, which is exponential in time complexity.
- Considerably smaller key size can be used for ECC compared to RSA (for example, the security of 160-bit ECC is equivalent to that for 1024-bit RSA).
- There is a computational advantage to using ECC with a shorter key length than a comparably secure RSA.

Elliptic Curve Cryptography (ECC)

Comparison of security issues between RSA and ECC

RSA	ECC
Hard problem: $n = p \times q$, finding $d \equiv e^{-1} \pmod{\phi(n)}$ is a hard problem given n and e .	Hard problem: $Q = kP$, where P and $Q \in E_p(a, b)$ are known, to compute k is a hard problem.